

Programming Docker Containers and Kubernetes Clusters on Real Cloud IBM Bluemix

A container image is a lightweight, standalone, executable package of a piece of software that includes everything needed to run it — code, runtime and system tools. Kubernetes is an open source system for automating deployment, scaling and managing containerised applications. Both these technologies have a meeting point in IBM Bluemix, which is a cloud Platform-as-a-Service (PaaS) developed by IBM.



Issues related to overheads and resource consumption due to increased use of virtualisation technology are being researched since a long time. The key limitations associated with virtualisation include upfront costs, security, software licensing and the complexities. In traditional cloud implementations, virtual machines that have fully dedicated operating systems along with separate memories are used. But these are not effective in the distributed high-performance computing environment.

Therefore, the need to work on a new technology like Docker containers arises. Using Docker technology, the containers are executed in association with the Docker engine rather than the core hypervisor.

Docker containers are quite effective and have enormous potential compared to traditional virtual machines (VMs). They can work on a single kernel with the sharing of multiple libraries, which is not possible in the typical VM based environment.

Container technology is very swift compared to traditional VMs and consumes minimum resources. Users can share a common library as well as the kernel to reduce the overhead on multiple modules. In addition, Docker containers are loaded and launched within a few seconds compared to the lengthier startup time for typical VMs.

Figure 3 depicts the integrated or the hybrid approach of traditional virtualisation and Docker container technology for specific applications. Using this approach, the cross-platform hybrid cloud can be programmed for high performance computing applications.

Working with containers on Real Cloud IBM Bluemix

IBM Bluemix is one of the powerful clouds in the Platform-as-a-Service (PaaS) segment. It supports enormous cloud based services including mobile applications, NoSQL databases, Internet of Things (IoT), network services,